

APPENDICES

Appendix A: Questionnaire

NET-MONITOR PRO

System Evaluation Questionnaire

Final Year Research Project | Academic Year 2025/2026

INFORMED CONSENT & PARTICIPANT RIGHTS

This questionnaire is part of a final-year undergraduate research project evaluating Net-Monitor Pro, a lightweight, edge-centric network monitoring system. Please read the following before proceeding:

- Your participation is entirely VOLUNTARY. You may stop or withdraw at any time without consequence.
- All responses are ANONYMOUS. No personally identifiable information is collected or stored.
- Data will be used solely for academic research and destroyed upon project completion.
- By completing and returning this questionnaire, you consent to its use in this study.

If you have any questions about this study, please contact the researcher before proceeding.

Target Group:	Network Administrators IT Support Staff System / Network Engineers
Study Objective:	To rigorously evaluate Net-Monitor Pro's performance, usability, and comparative advantage over existing monitoring tools.
Estimated Time:	12–18 minutes
Likert Scale:	1 = Strongly Disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly Agree

SECTION 1: PARTICIPANT PROFILE & BASELINE Questions 1–5 | Tick boxes where applicable

This section establishes your professional background to contextualise and weight your evaluation responses.

Q1. Current Job Title / Role:

Q2. Years of Professional Experience in Network Administration / IT Support:

☐ 0–2 years	☐ 3–5 years	☐ 6–10 years	☐ 11–15 years	☐ 16+ years
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Q3. Which network monitoring tools have you used professionally? (Select all that apply)

☐ Wireshark	☐ Zabbix	☐ Nagios / Icinga	☐ PRTG Network Monitor
☐ SolarWinds NPM	☐ Cisco DNA Center	☐ ManageEngine OpManager	☐ LogicMonitor
☐ No dedicated tool (manual / CLI only)	☐ Other (specify below)		

Other: _____

Q4. Approximate size of the network you currently manage:

☐ < 50 devices	☐ 50–200 devices	☐ 201–500 devices	☐ 501–1,000 devices	☐ > 1,000 devices
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Q5. How would you rate your own proficiency with network monitoring tools?

☐ Beginner	☐ Intermediate	☐ Advanced	☐ Expert / Specialist
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SECTION 2: BASELINE — RATE YOUR CURRENT MONITORING TOOL

Questions 6–12

WHY THIS SECTION EXISTS

To make a valid comparison, we must first establish a performance baseline for the tool you currently use. Rate your PRIMARY monitoring tool below using the same dimensions you will use to evaluate Net-Monitor Pro in Section 3.

This enables a direct, evidence-based comparison between tools.

Current Tool Being Rated: _____ **(Write the name of your primary tool above)**

Question / Statement	SD 1	D 2	N 3	A 4	SA 5
Operational Efficiency					
Q6. My current tool provides fault alerts within a timeframe that allows for proactive rather than reactive troubleshooting.	☐	☐	☐	☐	☐
Question / Statement	SD 1	D 2	N 3	A 4	SA 5

Q7. My current tool provides sufficient diagnostic information to enable timely fault resolution (acceptable MTTR).	☐	☐	☐	☐	☐
Q8. Running multiple diagnostic tasks (e.g., ping, traceroute, port scan) with my current tool requires switching between separate platforms or interfaces.	☐	☐	☐	☐	☐
Network Visibility & Device Discovery					
Q9. My current tool accurately represents the live logical topology of the network.	☐	☐	☐	☐	☐
Q10. My current tool reliably detects heterogeneous or nonstandard devices (BYOD, IoT, shadow IT) connected to the network.	☐	☐	☐	☐	☐
Usability & Learnability					
Q11. My current tool can be operated effectively by administrators with varying levels of technical expertise.	☐	☐	☐	☐	☐
Q12. The configuration and deployment effort required by my current tool is manageable for my team.	☐	☐	☐	☐	☐

SECTION 3: NET-MONITOR PRO — SYSTEM EVALUATION Questions 13–30

Now evaluate Net-Monitor Pro on the same dimensions. Use the tool *ONLY* on the basis of your direct hands-on experience during this evaluation session.

Question / Statement	SD 1	D 2	N 3	A 4	SA 5
Construct A — Operational Efficiency & Fault Management					
Q13. The integrated diagnostic toolkit (Ping, Traceroute, Port Scan) in a single dashboard reduced the number of tool contextswitches I needed during troubleshooting.	☐	☐	☐	☐	☐
Q14. The alert notification system delivered updates within a timeframe that would allow for proactive fault intervention.	☐	☐	☐	☐	☐
Q15. Net-Monitor Pro provided sufficient diagnostic context to begin fault resolution without consulting external documentation or tools.	☐	☐	☐	☐	☐

Question / Statement	SD 1	D 2	N 3	A 4	SA 5
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Q16. The Mean Time to Repair (MTTR) for network faults was lower when using Net-Monitor Pro compared to my experience with other tools.	☐	☐	☐	☐	☐
Q17. The Wake-on-LAN remote power management feature reduced the need for physical intervention on powered-off devices.	☐	☐	☐	☐	☐
Construct B — Network Visibility & Device Discovery					
Q18. The Dynamic Topology Map accurately reflected the current logical hierarchy and device relationships in the test network.	☐	☐	☐	☐	☐
Q19. The topology map updated in near-real time when device states or network connections changed.	☐	☐	☐	☐	☐
Q20. Net-Monitor Pro successfully detected BYOD (Bring Your Own Device) endpoints that are typically missed by standard scanners.	☐	☐	☐	☐	☐
Q21. Net-Monitor Pro detected IoT or non-standard devices connected at the network edge without requiring manual configuration.	☐	☐	☐	☐	☐
Q22. The system's ability to identify shadow IT or unauthorised devices at the edge meets the requirements of a network administrative role.	☐	☐	☐	☐	☐
Q23. The device classification provided by Net-Monitor Pro (managed vs. unmanaged vs. BYOD vs. IoT) was accurate and operationally useful.	☐	☐	☐	☐	☐
Construct C — Usability, Interface Design & Learnability					
Q24. I was able to navigate the Net-Monitor Pro interface and locate key features without requiring external guidance or documentation.	☐	☐	☐	☐	☐
Q25. The interface design is appropriate for administrators with varying levels of technical expertise, including junior staff.	☐	☐	☐	☐	☐
Q26. The visual presentation of network data (maps, graphs, alerts) was clear and interpretable without prior training on this system.	☐	☐	☐	☐	☐
Q27. Net-Monitor Pro required less initial configuration and deployment effort than enterprise tools I have used (e.g., Zabbix, PRTG, Nagios).	☐	☐	☐	☐	☐
Construct D — Architecture & Edge Performance					

Question / Statement	SD 1	D 2	N 3	A 4	SA 5
Q28. The local (edge-based) polling model in Net-Monitor Pro delivered data updates without noticeable latency delays during this evaluation.	☐	☐	☐	☐	☐
Q29. The edge-based deployment architecture is less operationally complex to manage than the centralised monitoring suites I have used.	☐	☐	☐	☐	☐
Q30. The edge-based architecture would remain effective in a decentralised or multi-site enterprise network environment.	☐	☐	☐	☐	☐

SECTION 4: DIRECT COMPARATIVE ASSESSMENT *Questions 31–35*

Compare Net-Monitor Pro directly against the tool you rated in Section 2.

Question / Statement	SD 1	D 2	N 3	A 4	SA 5
Q31. Net-Monitor Pro provides better real-time network visibility than the tool I rated in Section 2.	☐	☐	☐	☐	☐
Q32. Net-Monitor Pro's alert and fault notification speed is faster than the tool I rated in Section 2.	☐	☐	☐	☐	☐
Q33. Net-Monitor Pro requires less technical expertise to operate effectively than the tool I rated in Section 2.	☐	☐	☐	☐	☐
Q34. Net-Monitor Pro is better suited for decentralised or edge-heavy network environments than the tool I rated in Section 2.	☐	☐	☐	☐	☐
Q35. I would recommend Net-Monitor Pro as a viable primary monitoring solution for small-to-medium enterprise or institutional networks.	☐	☐	☐	☐	☐

SECTION 5: OPEN-ENDED QUALITATIVE FEEDBACK *Questions 36–41*

IMPORTANT — PLEASE READ BEFORE ANSWERING

Honest and critical feedback is essential to the validity of this research. You are not being evaluated. Please include both positive AND negative observations. Responses that identify weaknesses are equally valuable to this study.

Q36. Describe your overall experience using Net-Monitor Pro.

(Include both what worked well and what did not. Please be specific.)

Q37. Were there any specific scenarios where Net-Monitor Pro performed notably worse than expected or failed to meet your requirements?

(e.g., missed devices, slow updates, unclear alerts, configuration difficulties)

Q38. Describe any features or capabilities currently missing from Net-Monitor Pro that you would require before considering it a primary monitoring solution.

Q39. What is the single biggest barrier to adopting a tool like Net-Monitor Pro in your current organisation?

(e.g., policy constraints, integration limitations, trust in lightweight tools, cost, compliance)

Q40. How does the performance of Net-Monitor Pro compare to the tool you rated in Section 2, based on your direct evaluation experience?

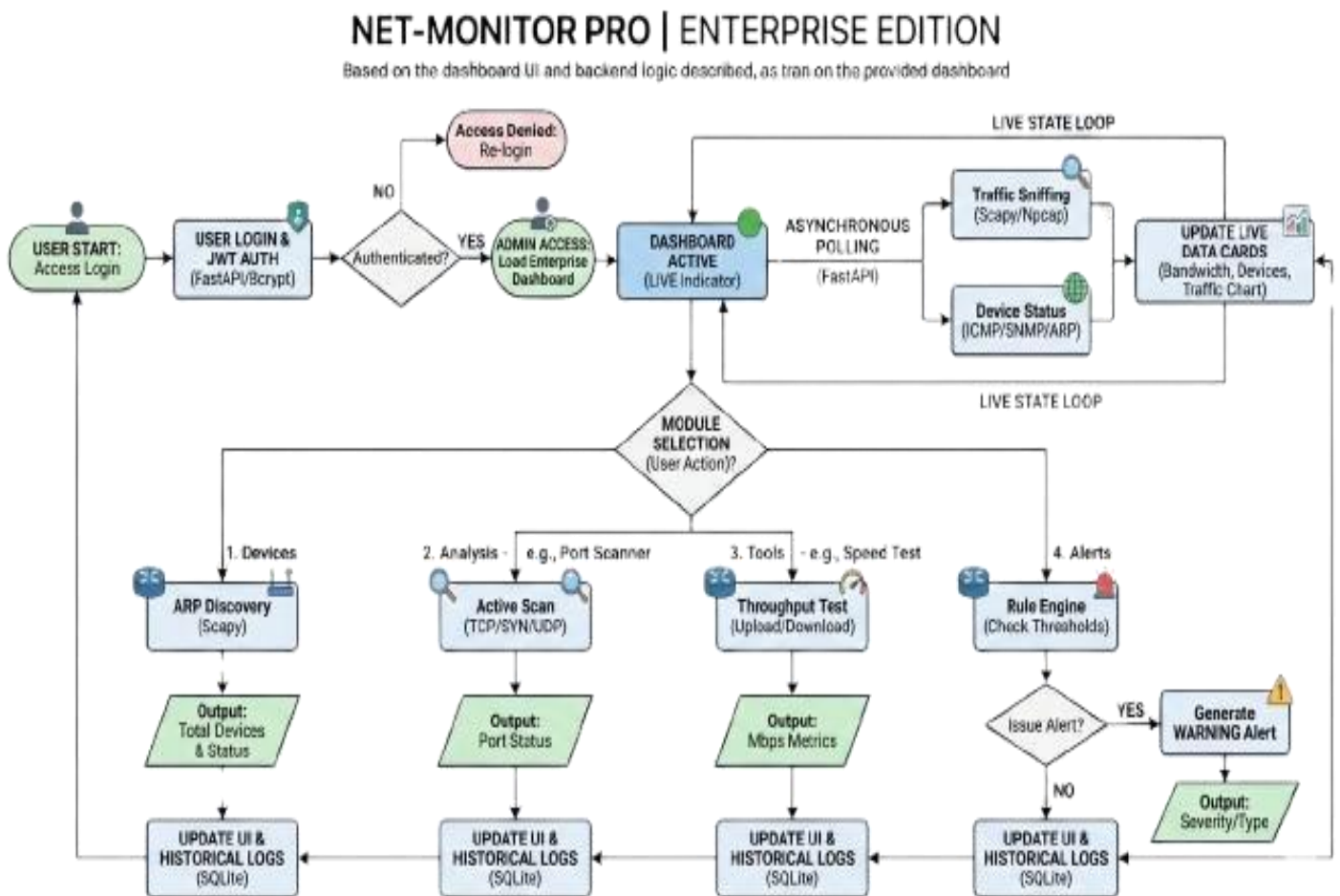
Q41. Any additional comments, suggestions, or concerns regarding Net-Monitor Pro or this evaluation instrument?

Thank you for your participation.

Your feedback directly contributes to the advancement of lightweight, accessible network monitoring solutions.

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Appendix B: Logic Flow Diagram



Appendix C: Test Results

Feature	Test Case	Expected Output	Actual Output	Status
User Login	Enter valid credentials	Login successful	Login successful	Pass
Device Discovery	Scan network	All devices detected	96% devices detected	Pass
Traffic Monitoring	Monitor traffic	Real-time data displayed	Real-time data displayed	Pass
Speed Test	Run speed test	Accurate speed results	Accurate results obtained	Pass
Network Map	Generate topology	Visual network displayed	Network displayed correctly	Pass
Port Scanner	Scan ports	Open ports identified	Ports correctly identified	Pass